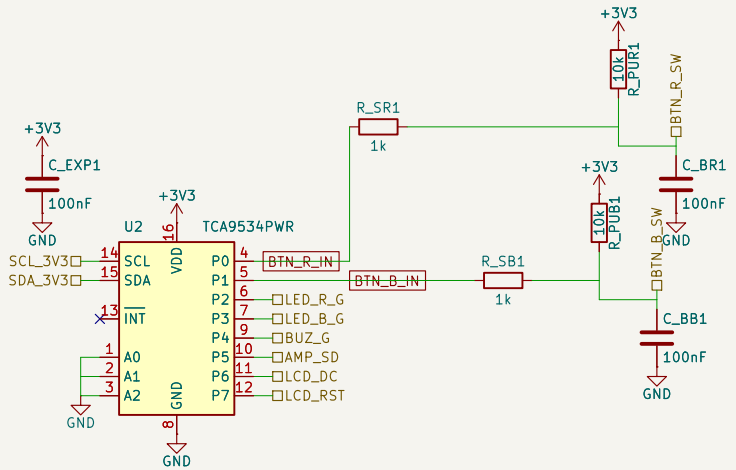
## Heltec WiFi LoRa 32 V4 (plugs into J2/J3)

#X_PACK
Pack board (P5)
XH-6
J_PWR1
PWR
VBAT_SW
SDA_3V3
SCL_3V3
C_BULK1 47uF 10V
C_BULK2 47uF 10V
C_5V1 10uF
C_5V2 100nF
PWR_FLAG
J2
Heltec_J2
J3
Heltec_J3
R_SDA3 4K7
R_SCL3 4K7
#X_HBAT
Heltec V4 battery
XH-2 to the module's SH1.25
J_HBAT1
HELTEC_BAT
J_I2C1
I2C
TP_3V3
TP_5V
TP_VSW
TP_GND
H1 M3
H2 M3
H3 M3
H4 M3
I2S DAC and speaker amp
I2S_BCLK
I2S_LRCLK
I2S_DIN
AMP_SD
SPK_NC
SPK_P
AMP_SD
LCD_MOSI
LCD_SCK
J_LCD1
LCD
R_BLPD1 10k
LCD_RST
LCD_DC
LCD_BL
#X_LCD
3.5" ILI9488 SPI TFT
XH-8, straight
GPIO expander (0x20), button inputs
SDA_3V3
SCL_3V3
AMP_SD
LCD_RST
LCD_DC
BTN_R_SW
BTN_B_SW
BUZ_G
LED_B_G
LED_R_G
Low-side drivers: button LEDs, buzzer
BTN_R_LEDK
BUZ_G
LED_B_G
LED_R_G
BTN_B_LEDK
BUZ_K
1N4148W
D_FLY1
J_NFC1
NFC
C_NFC1 100uF 6.3V
#X_NFC
PN532 (I2C)
XH-4
J_BTN_R1
BTN_R
BTN_R_SW
BTN_R_LEDK
#X_BTN_R
ONPOW LAS1-AGQ-11E/R/6V
XH-4
J_BTN_B1
BTN_B
BTN_B_SW
BTN_B_LEDK
#X_BTN_B
ONPOW LAS1-AGQ-11E/B/6V
XH-4
J_BUZ1
BUZ
BUZ_K
#X_BUZ
BZ-38 buzzer
XH-2
J_SPK1
SPK
SPK_N
SPK_P
#X_SPK
VISATON K 50, 8 ohm
XH-2, BTL: neither side is GND

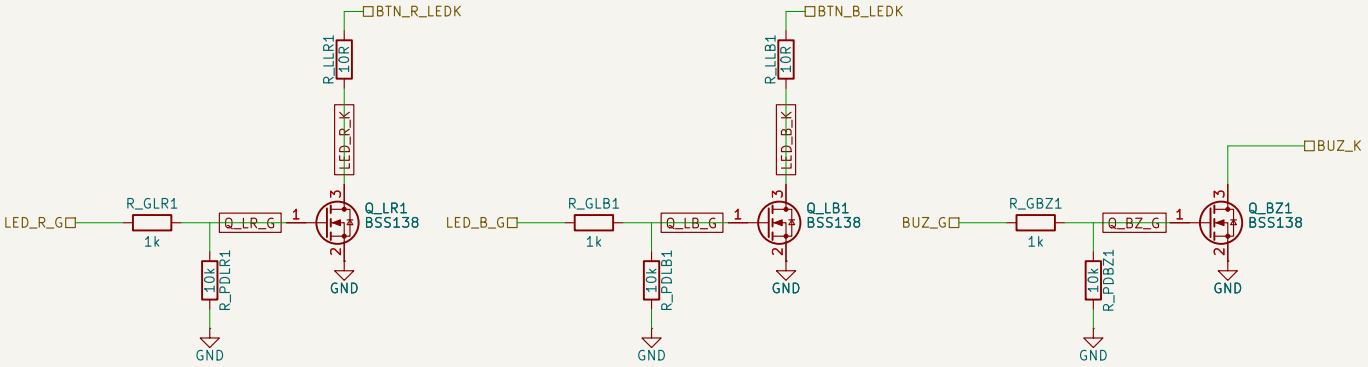
Sheet: /
File: carrier_tme.kicad_sch
**Title: AirsoftCounter v2 carrier (hand-build): system**
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 1/4

## GPIO expander (0x20), button inputs



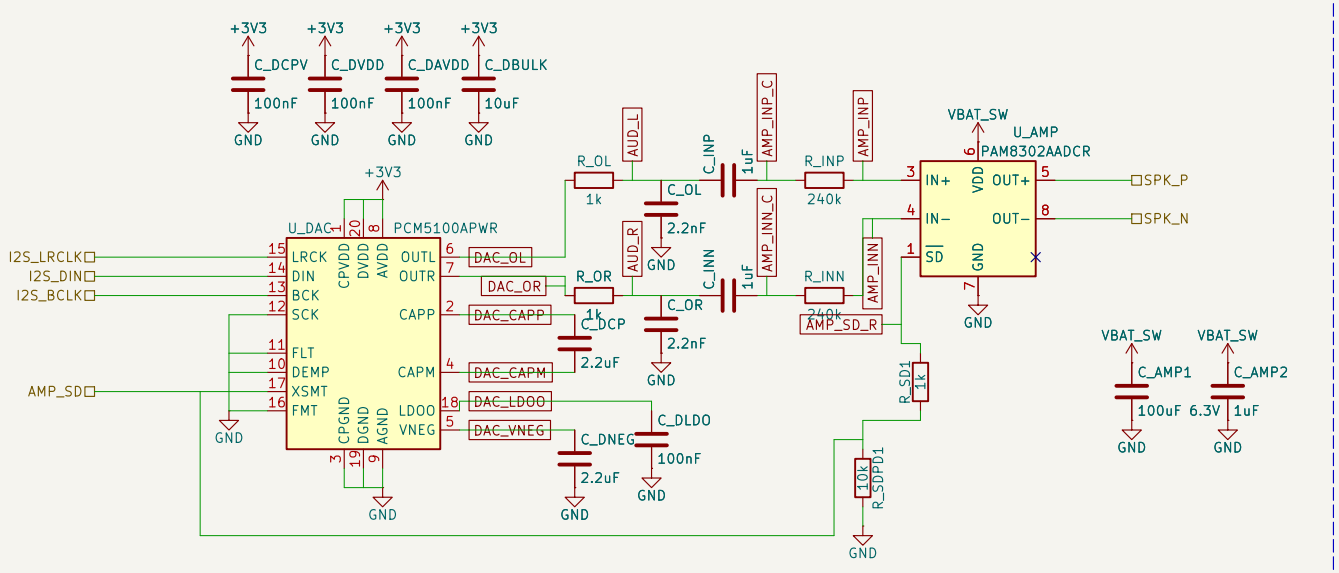

| | | |
|---|---|---|
| Sheet: /GPIO expander (0x20), button inputs/ | | |
| File: carrier_tme-expander.kicad_sch | | |
| **Title: AirsoftCounter v2 carrier (hand-build): GPIO expander (0x20), button i** | | |
| Size: A5 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.7 | | Id: 2/4 |

## Low-side drivers: button LEDs, buzzer




| | | |
|---|---|---|
| Sheet: /Low-side drivers: button LEDs, buzzer/ | | |
| File: carrier_tme-drivers.kicad_sch | | |
| **Title: AirsoftCounter v2 carrier (hand-build): Low-side drivers: button LEDs,** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.7 | | Id: 3/4 |

## I2S DAC and speaker amp




| | | |
|---|---|---|
| Sheet: /I2S DAC and speaker amp/ | | |
| File: carrier_tme-audio.kicad_sch | | |
| **Title: AirsoftCounter v2 carrier (hand-build): I2S DAC and speaker amp** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.7 | | Id: 4/4 |